

KSHITIJ BACHHAV

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Education

IIIT Nagpur

B.Tech in Internet of Things (IoT)

Nagpur, Maharashtra

2023 – 2027

Technical Skills

Languages: JavaScript, TypeScript, C++, Python

Frontend: React.js, Tailwind CSS

Backend: Node.js, Express.js, FastAPI

Databases: PostgreSQL, MongoDB, Prisma ORM, Redis

Tools: Git, GitHub, Docker, Postman

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, DBMS, RESTful APIs, Authentication(JWT, RBAC, OAuth), Operating Systems, Computer Networks

Projects

Multi-Tenant Project Management SaaS | *React.js, Express.js, PostgreSQL, Docker* | [GitHub](#) | [Live](#) **2025 – Present**

- Architected a multi-tenant SaaS collaboration platform with strict data isolation across **4 hierarchical layers** (organizations, workspaces, projects, tasks), enabling scalable onboarding for teams of any size.
- Secured **100% of REST API endpoints** by implementing a robust authentication flow using JWT access tokens, refresh-token rotation, bcrypt-hashed storage, and token-reuse detection, with **rate-limited routes** to prevent abuse.
- Enforced organization-scoped **RBAC** (admin, member, viewer) across backend middleware and React Router guards, covering invitations, member lifecycle management, and protected onboarding flows across **3 role tiers**.
- Delivered a full Kanban workflow with drag-and-drop reordering, optimistic UI updates, persisted task ordering, server-side filtering, cursor-based pagination, soft deletes, and audit activity logs backed by **indexed PostgreSQL tables** with unique constraints.

URL Shortener | *Node.js, Express.js, Prisma ORM, Redis, Docker* | [GitHub](#) **2025**

- Engineered a URL shortening service with Node.js, Express, and Prisma ORM over PostgreSQL, exposing REST endpoints for link creation and resolution that serve **302 redirects** via unique-indexed short-code lookups.
- Generated collision-resistant random short codes across a **62⁷** keyspace with retry-on-conflict insertion, avoiding the enumeration and count-leakage risks of sequential ID encoding.
- Added a Redis cache-aside layer for hot short codes, achieving **sub-millisecond** redirect lookups with a graceful fallback to PostgreSQL if the cache is unavailable.
- Hardened the API with JWT authentication and per-client rate limiting (**200 redirects/min, 30 creates/15min**), and containerized the full stack with Docker for reproducible deployment.

Policast | *React, FastAPI, PostgreSQL, Websockets, Groq API, GDELT* | [GitHub](#) **December 2025**

- Developed a real-time geopolitical prediction market platform with **22 REST API endpoints** across markets, signals, drivers, and scheduler control, tracking 52 live prediction markets with full CRUD and admin-gated mutation routes.
- Engineered a multi-provider **LLM forecasting pipeline** (**Groq Llama-3.3-70B** with Claude and GPT-4o automatic fallback) ingesting **GDELT** and RSS news signals every 2 hours, recalculating probabilities every 4 hours with guardrails clamping shifts to **±8% per cycle**.
- Integrated a **WebSocket** broadcast layer with a custom connection manager and heartbeat protocol to push live probability updates to all connected clients, eliminating client-side polling across the React frontend.
- Designed an on-demand AI briefing feature using **FastAPI Server-Sent Events** to stream structured **180-230 word** geopolitical analyses token-by-token to React, with cache invalidation triggered on probability changes.

Achievements

Winner (1st Place): Technex GameJam 2024, IIT BHU – ranked **1st** among all participating teams in a national-level game development hackathon.

DSA: Solved **200+ problems** across LeetCode and GeeksForGeeks, covering arrays, trees, graphs, DP, and greedy algorithms.